

DTGB: A Benchmark Suite for Dynamic Text-Attributed Graphs

DTGB (Zhang et al., 2024) introduces the first comprehensive benchmark specifically designed for **Dynamic Text-Attributed Graphs (DyTAGs)** — graphs where both structure and **natural language attributes** evolve over time. It offers eight large-scale datasets from domains like e-commerce, email, knowledge graphs, and multi-round dialogue, each annotated with **node and edge text**, timestamps, and edge types. DTGB evaluates how well **dynamic graph models and LLMs** handle temporal, structural, and textual evolution simultaneously.

Dynamic Text-Attributed Graph Construction (With Examples)

DTGB graphs are constructed from raw datasets like Amazon, Yelp, StackExchange, and Enron by associating each interaction with a timestamp, edge category, and free-text descriptions.

Example (Amazon movies):

- **Node text:**
Movie: *"The Godfather"*
Description: *"An organized crime dynasty's aging patriarch transfers control of his empire to his reluctant son."*
- **Edge text (user review):**
"One of the best movies I've ever seen. Acting and direction are top-notch."
 - Timestamp: T = March 2021
 - Edge category: (Rating = 5)

These interactions are segmented into discrete time intervals, and each subgraph snapshot captures both textual and structural dynamics at that time.

LLM-Based Textual Relation Generation (With Examples)

DTGB goes beyond link prediction and node classification by proposing a new task: **textual relation generation**.

Given the descriptions of two nodes and their past interactions, an LLM (e.g., GPT-4, LLaMA-3) is prompted to generate the **text of their future interaction**.

Example (StackExchange dialogue):

- Node A: "How to install Python on Ubuntu?"
- Node B: "You can use apt-get or download Anaconda."

Historical Context (previous Q&A threads):

- A: “I’m new to Linux”
- B: “Use `sudo apt-get install` for most packages”

LLM prompt:

“Based on the historical interactions and descriptions, generate the future message from A to B.”

LLM Output:

“Thanks! I installed it with apt-get and it works great. Any tips for setting up a virtual environment?”

This setup evaluates LLMs’ ability to **reason over temporal interaction graphs** enriched with text.

Benchmark Tasks and LLM Integration

DTGB defines four core tasks:

1. **Future link prediction** (predict future edges)
2. **Destination node retrieval** (rank likely target nodes)
3. **Edge classification** (predict relation types like rating or sentiment)
4. **Textual relation generation** (generate the actual edge text)

Text attributes are encoded with **BERT embeddings**, and used to initialize dynamic GNNs like TGAT, DyRep, DyGFormer.

For generation, LLMs like GPT-3.5, GPT-4o, LLaMA-3, and Vicuna are prompted with structured historical graph data.

Results show that:

- LLMs benefit from supervised fine-tuning (SFT) on graph-text pairs.
- Bert-based initializations improve dynamic GNN performance, especially in inductive settings.
- Current LLMs struggle with **long-range structural reasoning**, motivating better graph-aware prompting or tokenization strategies.

Why It Matters for the Survey

DTGB fills a crucial gap by providing a **standardized testbed** to evaluate how well LLMs and

dynamic GNNs can jointly model evolving graphs and language. It doesn't propose a new model, so it **does not compete** with research that builds **new methods for LLM+dynamic graph integration**.

Instead, it enables such research by offering datasets, tasks, and baselines — making it a foundational resource for any future work aiming to improve **temporal reasoning over graph-structured text**.